

MASTERCLASS

Linear Algebra for ML

The maths behind neural networks — made visual

 3 hours

 Intermediate

 Live

 Prasant Mishra

ABOUT THIS COURSE

Every AI model is built on linear algebra. This session builds a visual, geometric intuition for vectors, matrices, and transformations — exactly the parts that show up in ML.

Prerequisites: Basic algebra. Calculus helpful but not required.

WHAT YOU'LL LEARN

- 1 Understand matrix operations geometrically
- 2 Explain PCA and embeddings intuitively
- 3 Read ML papers without freezing at the maths
- 4 Use NumPy confidently for linear algebra

CURRICULUM · 3 SESSIONS

SESSION HOUR 1

Vectors & Spaces

What a vector really is

Vector operations — addition, scaling, dot product

Basis vectors and span

Linear independence

SESSION HOUR 2

Matrices as Transformations

Matrix multiplication as composition of transformations

Transpose, inverse

Determinant — what it actually means

Systems of equations

SESSION HOUR 3

Eigenvalues, PCA & Neural Nets

Eigenvalues & eigenvectors visually

PCA — how dimensionality reduction works

How weights in neural networks are matrices

Practical NumPy session

WHAT'S INCLUDED

- ✓ Full session recording (lifetime access)
- ✓ Visual cheat sheet PDF
- ✓ Learner's Certificate
- ✓ Live doubt clearing session (1 hr)
- ✓ NumPy code notebook

UPCOMING BATCHES



4 Jul 2026 · 3:00 PM IST



1 Aug 2026 · 3:00 PM IST



learner

₹799

Live · 3 hours · Recorded version available at ₹199

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YOUR INSTRUCTOR



Prasant Mishra

Finance professional with 14+ years across equity research, investment banking, and asset management. Passionate about making complex financial concepts accessible through real-world case studies and hands-on learning.

